

# VPA ICT — IGCSE Y10S1: Advanced Practical & Web Authoring

## IGCSE ICT (0417) — Year 10 Semester 1: Advanced Practical Skills

Syllabus: CIE IGCSE ICT (0417) Paper 3 | Exam: Paper 3 — Data Analysis, Web Authoring & Databases (2h30, 80 marks)  
Duration: 18 weeks | Lesson Length: 40 minutes

### Paper 3 Syllabus Coverage

| Skill Area                      | Syllabus Ref | Weeks   | Software                            |
|---------------------------------|--------------|---------|-------------------------------------|
| Data Analysis (Spreadsheets/DB) | 13.1 - 13.5  | 1 - 9   | Excel/Calc + Access/Base            |
| Web Authoring                   | 14.1 - 14.5  | 1 - 9   | Dreamweaver / Text Editor + Browser |
| Databases                       | 15.1 - 15.5  | 10 - 18 | Microsoft Access / LibreOffice Base |

### Semester Objectives

- By the end of Semester 1, students will be able to:
- ☐ Perform advanced data analysis using spreadsheets and databases
  - ☐ Create multi-page websites using HTML and CSS
  - ☐ Design and query relational databases
  - ☐ Integrate data between applications (web forms, database links)
  - ☐ Work with large datasets efficiently
  - ☐ Test and debug web pages across browsers

### Software Requirements

| Software                  | Version               | Alternative        | Exam Notes                                                |
|---------------------------|-----------------------|--------------------|-----------------------------------------------------------|
| Microsoft Access          | 2016+                 | LibreOffice Base   | Forms, queries, reports, relationships required           |
| Microsoft Excel           | 2016+                 | LibreOffice Calc   | Advanced formulas, pivot tables, what-if required         |
| Dreamweaver / Text Editor | Any                   | Notepad++, VS Code | HTML/CSS hand-coding tested; WYSIWYG not sufficient alone |
| Web Browsers              | Chrome, Firefox, Edge | —                  | Cross-browser testing essential                           |

### Paper 3 Exam Format

Duration: 2 hours 30 minutes

Marks: 80

Structure: Integrated tasks combining data analysis, web authoring, and databases

Evidence: Screen shots, printed pages, electronic files in designated folders

Key difference from Paper 2: Paper 3 requires INTEGRATION between applications (e.g., web form writes to database, spreadsheet data displayed on web page)

## Differentiation for Advanced Practical Skills

| Level  | Data Analysis                                               | Web Authoring                                                        | Databases                                                        |
|--------|-------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------|
| Bronze | Basic formulas and one chart                                | 2-page site with basic HTML                                          | Single table with simple queries                                 |
| Silver | Multiple functions, conditional formatting, what-if         | 3-page site with CSS, tables, images, links                          | Relationships, multi-table queries, basic forms                  |
| Gold   | Complex nested formulas, pivot tables, multiple chart types | 4+ page site with external CSS, forms, navigation, consistent design | Full relational design, parameterized queries, formatted reports |